Model p.Gašpara

$$\boxed{R_{th} = \frac{1}{a^2\pi\lambda}\int_0^\infty \frac{\sin^2(\xi a)\tanh(\xi L)}{\xi^3}\,d\xi}$$

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x}-1}{e^{2x}+1} = \frac{1-e^{-2x}}{1+e^{-2x}}$$

Upravím:

Zavedeme proměnné:

$$\eta = \xi L, \quad \xi = \frac{\eta}{L}, \quad p = \frac{a}{L}$$

Z toho plyne:

$$\xi a = p\eta, \quad \xi L = \eta, \quad pL = a$$

$$\boxed{R_{th} = \frac{L^2}{a^2\pi\lambda}\int_0^\infty \frac{\sin^2(p\eta)\tanh(\eta)}{\eta^3}d\eta, \quad \text{pro } p = a/L \ll 1}$$

Pro $p \ll 1$:

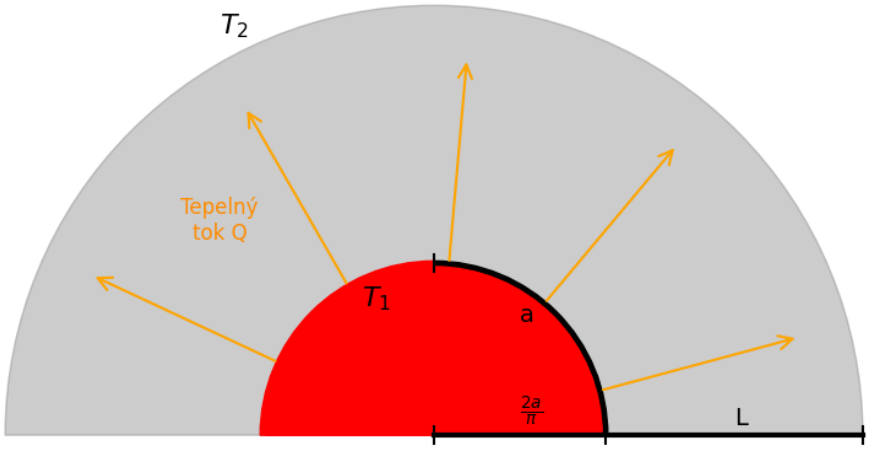
- Pokud $\eta < \frac{1}{p} \implies \sin(p\eta) \approx p\eta$
- Pokud $\eta > \frac{1}{p} \gg 1 \implies \tanh(\eta) \approx 1$

$$\begin{aligned}
R_{th} &\approx \frac{L^2}{a^2\pi\lambda}\left[\int_0^{1/p}\frac{(p\eta)^2\tanh(\eta)}{\eta^3}d\eta + \int_{1/p}^\infty \frac{\sin^2(p\eta)\cdot 1}{\eta^3}d\eta\right] \\
&= \frac{L^2p^2}{a^2\pi\lambda}\left[\int_0^{1/p}\frac{\tanh(\eta)}{\eta}d\eta\right] + \frac{L^2}{a^2\pi\lambda}\left[\int_{1/p}^\infty \frac{\sin^2(p\eta)}{\eta^3}d\eta\right] \\
&= \frac{L^2p^2}{a^2\pi\lambda}\left[\int_0^1\frac{\tanh(\eta)}{\eta}d\eta + \int_1^{1/p}\frac{1}{\eta}d\eta\right] + \frac{L^2p^2}{a^2\pi\lambda}\left[\int_1^\infty \frac{\sin^2(x)}{x^3}dx\right] \\
&\approx \frac{1}{\pi\lambda}\left[\int_0^1 \eta\,d\eta + \int_1^{1/p}\frac{1}{\eta}d\eta + \int_1^\infty \frac{\sin^2(x)}{x^3}dx\right] \\
&\approx \frac{1}{\pi\lambda}\left[[\ln(\eta)]_1^{1/p} + \mathcal{O}(1)\right] = \frac{1}{\pi\lambda}\left[1 + \mathcal{O}(1) + \ln\left(\frac{1}{p}\right) - \ln(1)\right]
\end{aligned}$$

$$\boxed{R_{th} \approx \frac{1}{\pi\lambda}\left[1 + \mathcal{O}(1) + \ln\left(\frac{L}{a}\right)\right] \approx \frac{1}{\pi\lambda}\left[1 + \ln\left(\frac{L}{a}\right)\right]}$$

Jiný hrubý model

Vodivá válcová tyčka o poloměru $\frac{2a}{\pi}$ v ose válce o méně vodivém prostředí o poloměru $\frac{2a}{\pi} + L$. Obojí má délku **b**. Teplota tyčky je konstatní T_1, teplota válce o polměru $(\frac{2a}{\pi} + L)$ na povrchu je T_2. Zde ekvivalent tloušťky málo vodivé desky je L.



$$Q = \frac{(T_2 - T_1)}{R_{th}} = q_1 \pi a b = q(r) \pi r b = q_2 \pi L b$$

$$q = -\lambda \frac{dT}{dr} = \frac{Q}{\pi r b} \quad \Longrightarrow \quad \int_{T_1}^{T_2} dT = -\frac{Q}{\pi b \lambda} \int_{\frac{2a}{\pi}}^{\frac{2a}{\pi}+L} \frac{dr}{r}$$

$$R_{th} = \frac{(T_2 - T_1)}{Q} = \frac{\ln\left(\frac{\frac{2a}{\pi}+L}{\frac{2a}{\pi}}\right)}{\pi b \lambda} = \frac{\ln\left(1 + \frac{\pi L}{2a}\right)}{\pi b \lambda}$$

$$\boxed{R_{th} = \frac{1}{\pi b \lambda} \ln(1 + \frac{\pi L}{2a}) \quad [K/W]}$$

Kontrola pro L«a:

$$R_{th} = \frac{\ln(1 + \frac{\pi L}{2a})}{\pi b \lambda} \approx \frac{\frac{\pi L}{2a}}{\pi b \lambda} = \frac{L}{2 \lambda a b}$$

oproti

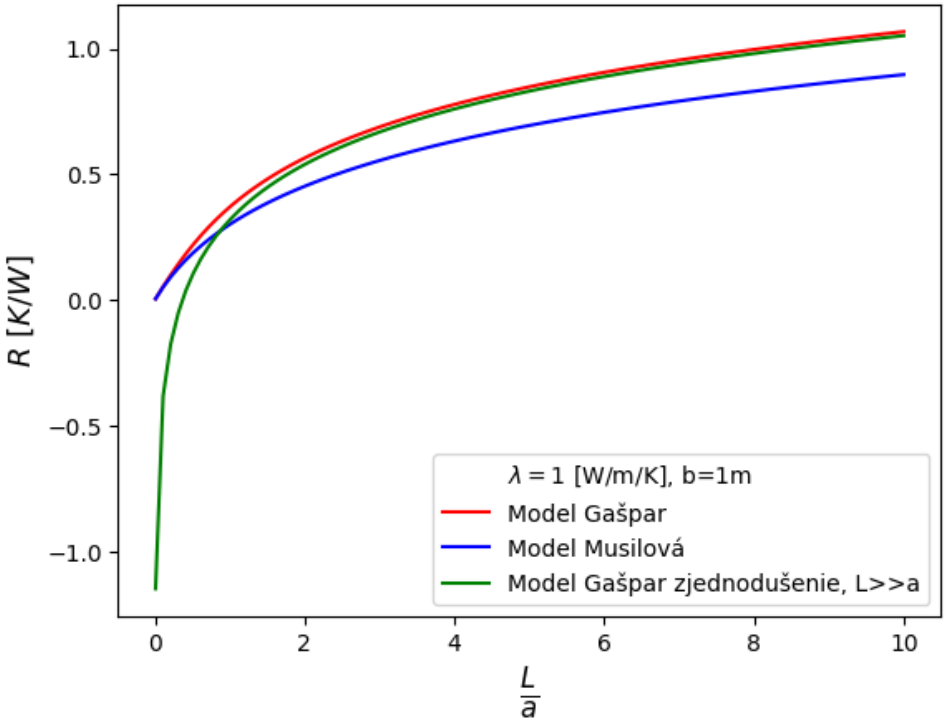
$$R_{th} = \frac{1}{2\lambda} \frac{L}{ab}$$

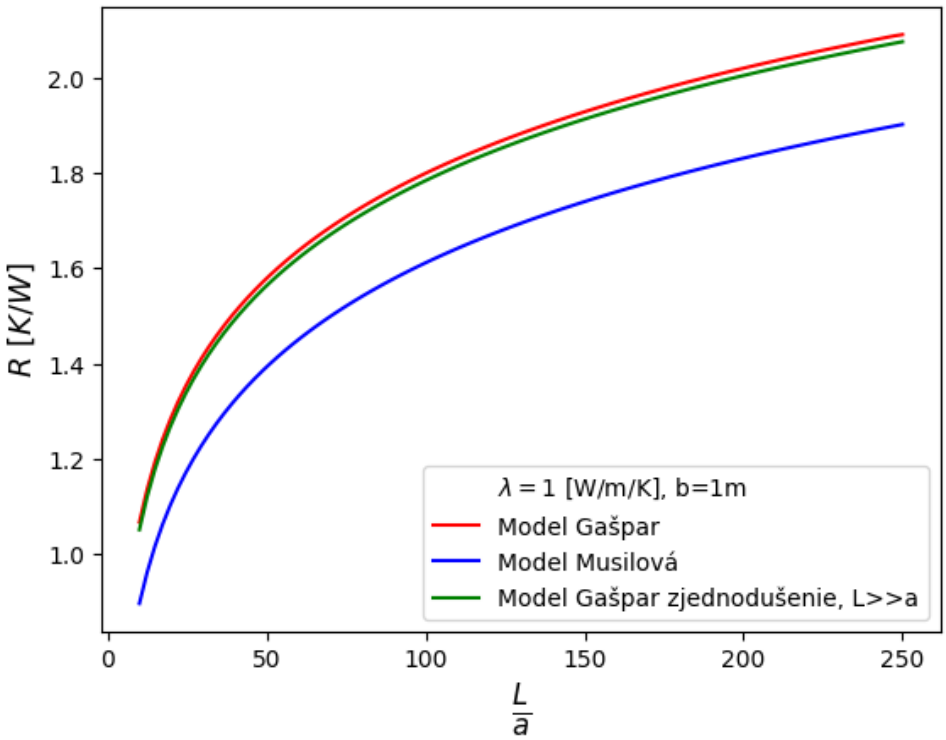
odvozenému panem Gašparem.

Poznámka:

v modelu s válci jsou vzorečky pro délku válce **b**, p.Gašpar má pro délku 1m.

Porovnanie modelov


1.0
0.5
0.0
−0.5
−1.0
R [K/W]
0
2
4
6
8
10
L/a
λ = 1 [W/m/K], b=1m
Model Gašpar
Model Musilová
Model Gašpar zjednodušenie, L>>a


2.0
1.8
1.6
1.4
1.2
1.0
R [K/W]
0
50
100
150
200
250
L/a
λ = 1 [W/m/K], b=1m
Model Gašpar
Model Musilová
Model Gašpar zjednodušenie, L>>a